

Linear Algebra – The Backbone of AI/ML

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Why study linear algebra ?

- Vectors and Matrices: Operations, properties, and applications in AI/ML.
- Eigenvalues and Eigenvectors: Principal Component Analysis (PCA).
- Vector Spaces: Concepts of basis, dimension, and linear independence.
- Matrix Decomposition: Singular Value Decomposition (SVD) and its applications.
- Gradient and Hessian: Introduction to vector calculus and its role in optimization

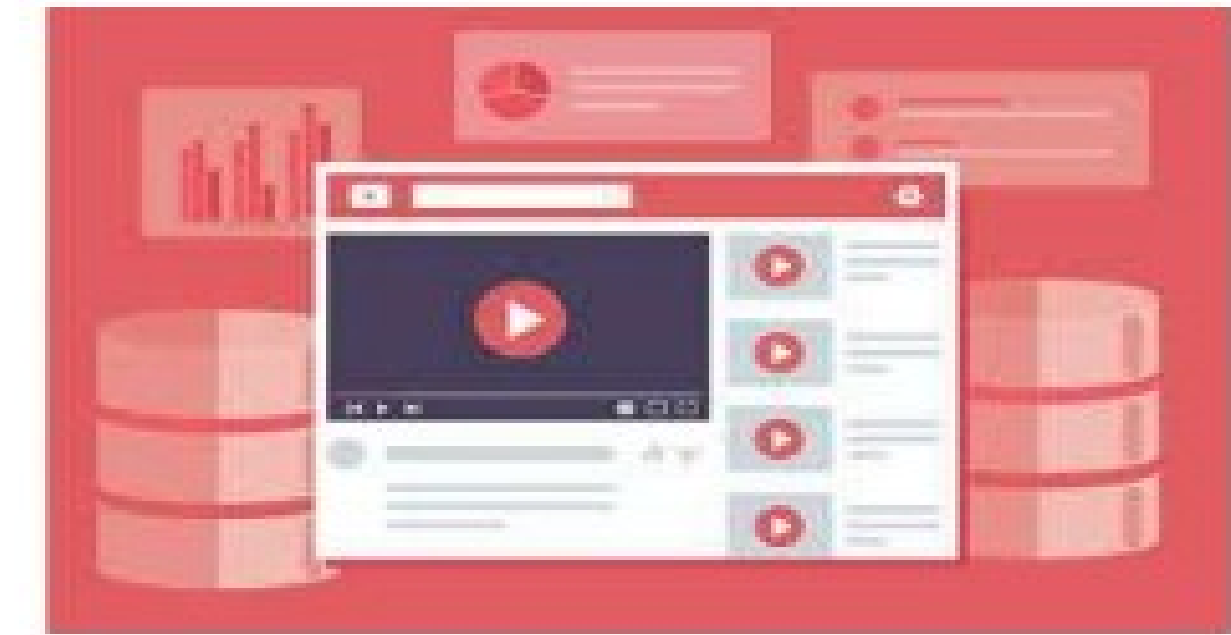
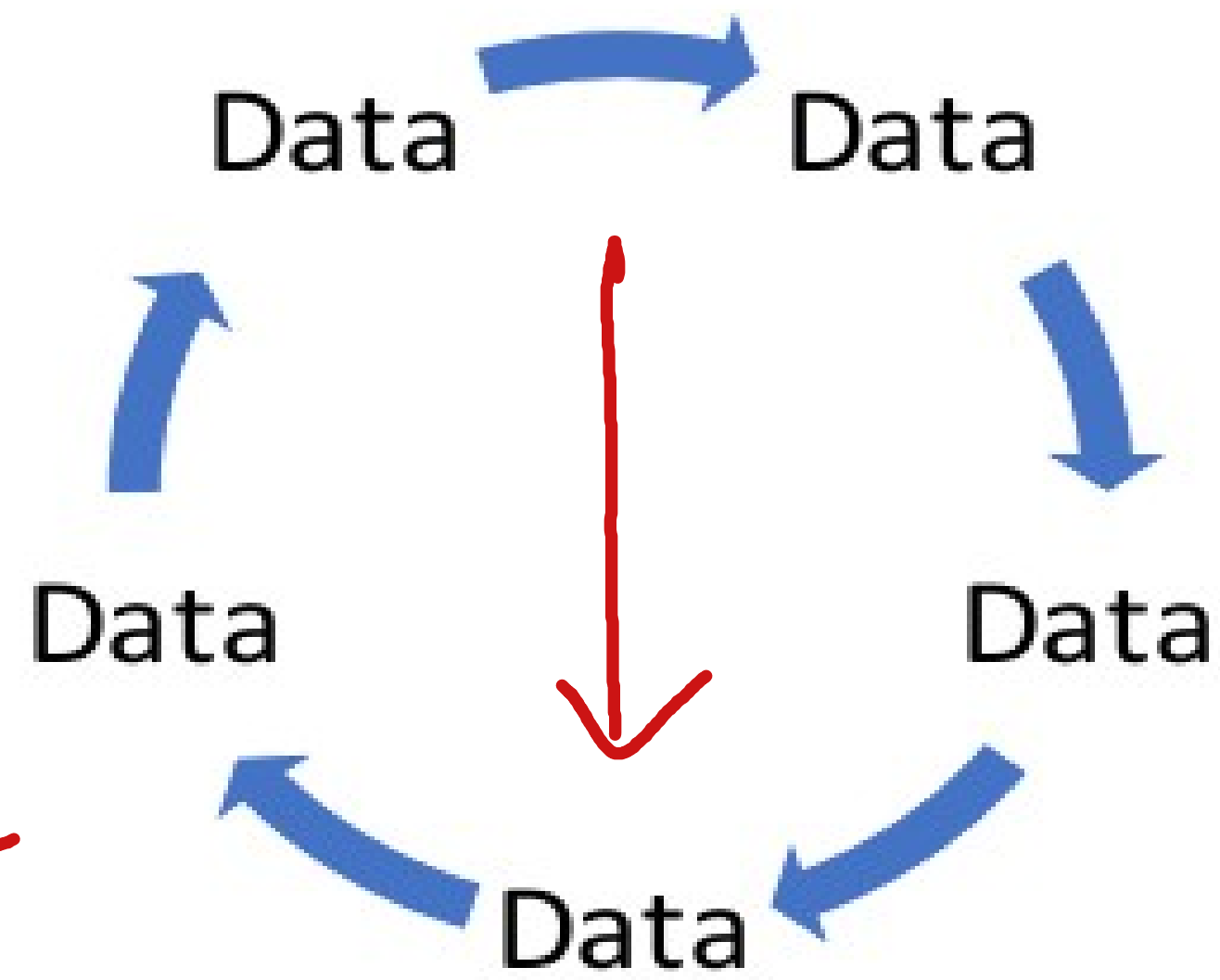


- Lecture Completion
- Summary
- Problem Solving for H/W

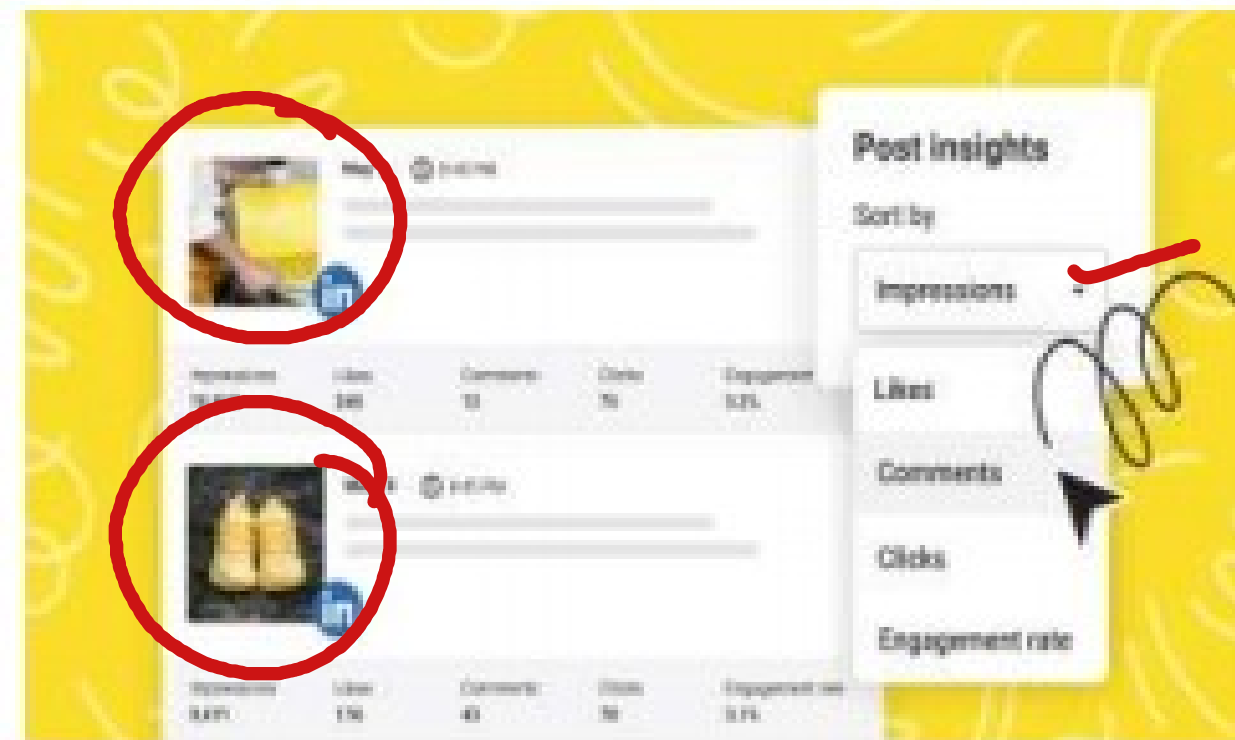


LinkedIn
post prince

→ Text data



video data



Text data

Main
Challenge

How to Make Use of this RAW Data

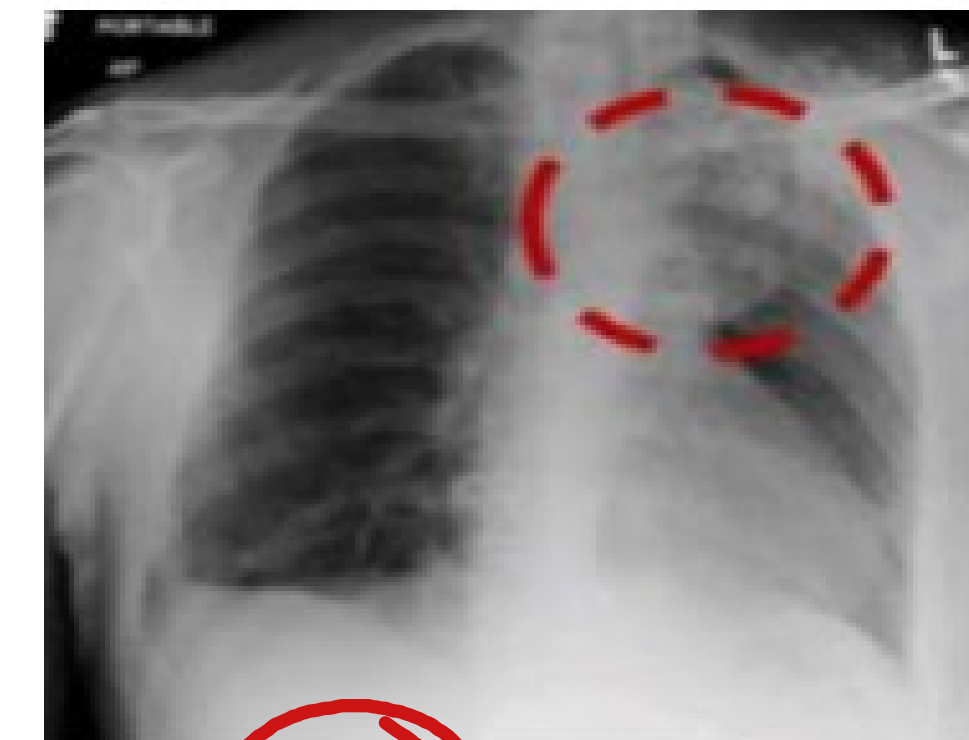
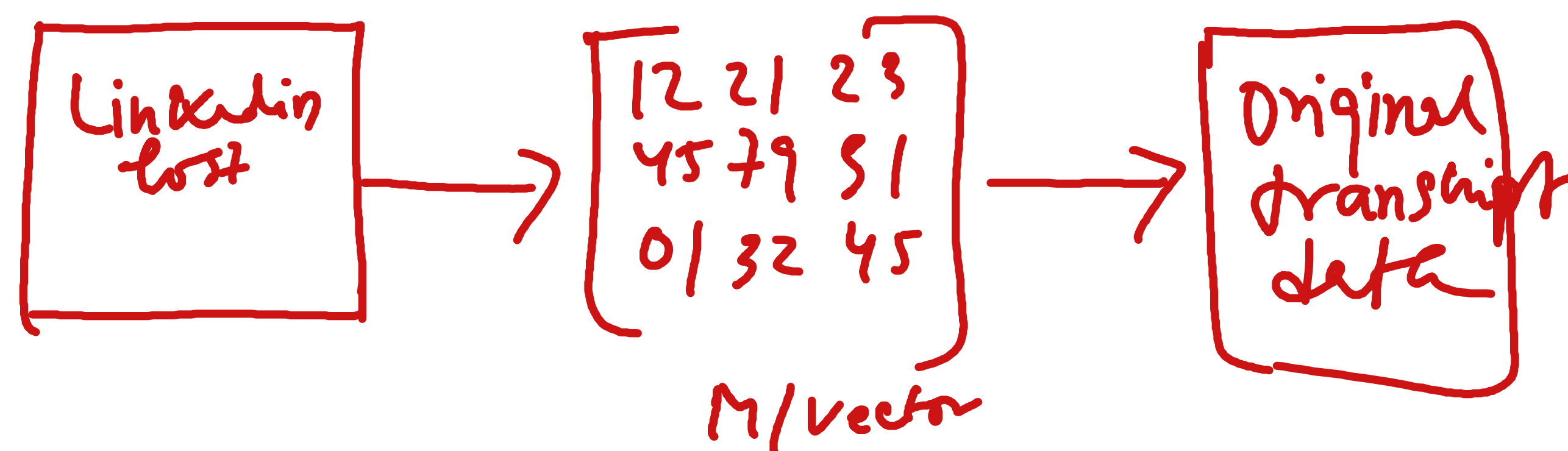
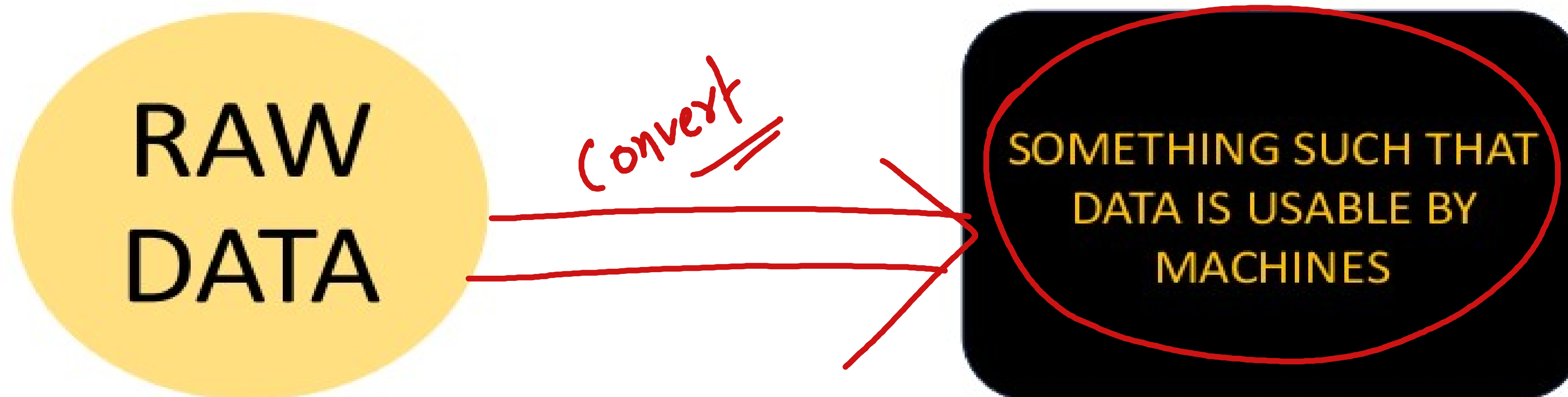


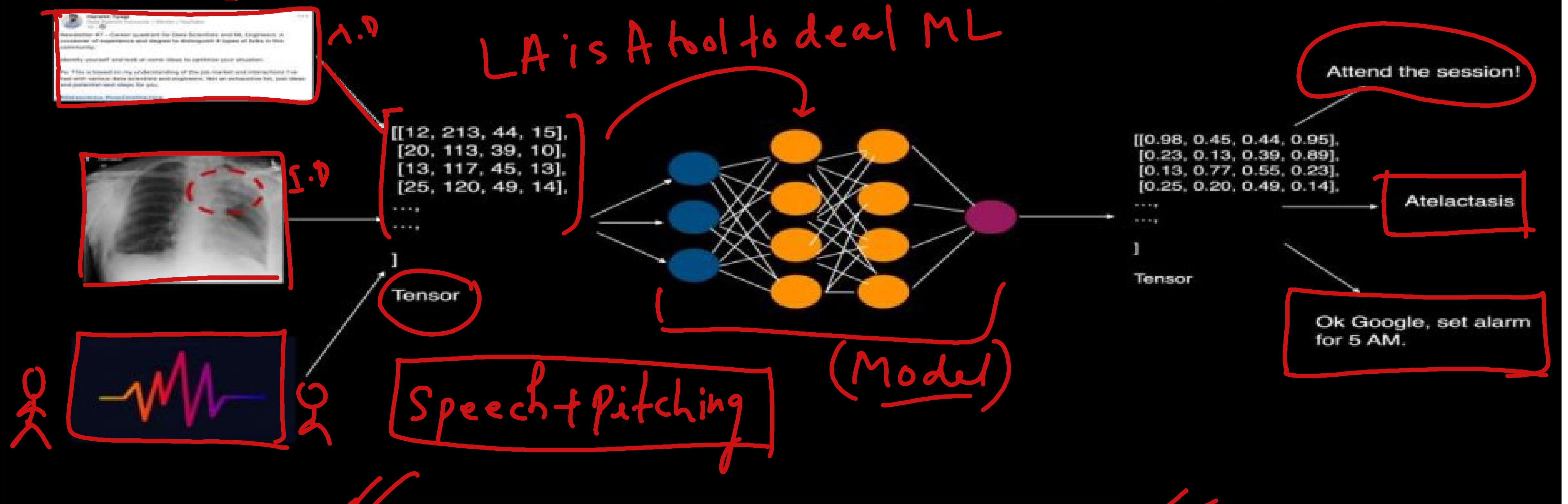
Image data

Pic





ML models Leveraging Linear Algebra



From Data to Vectors and Matrices



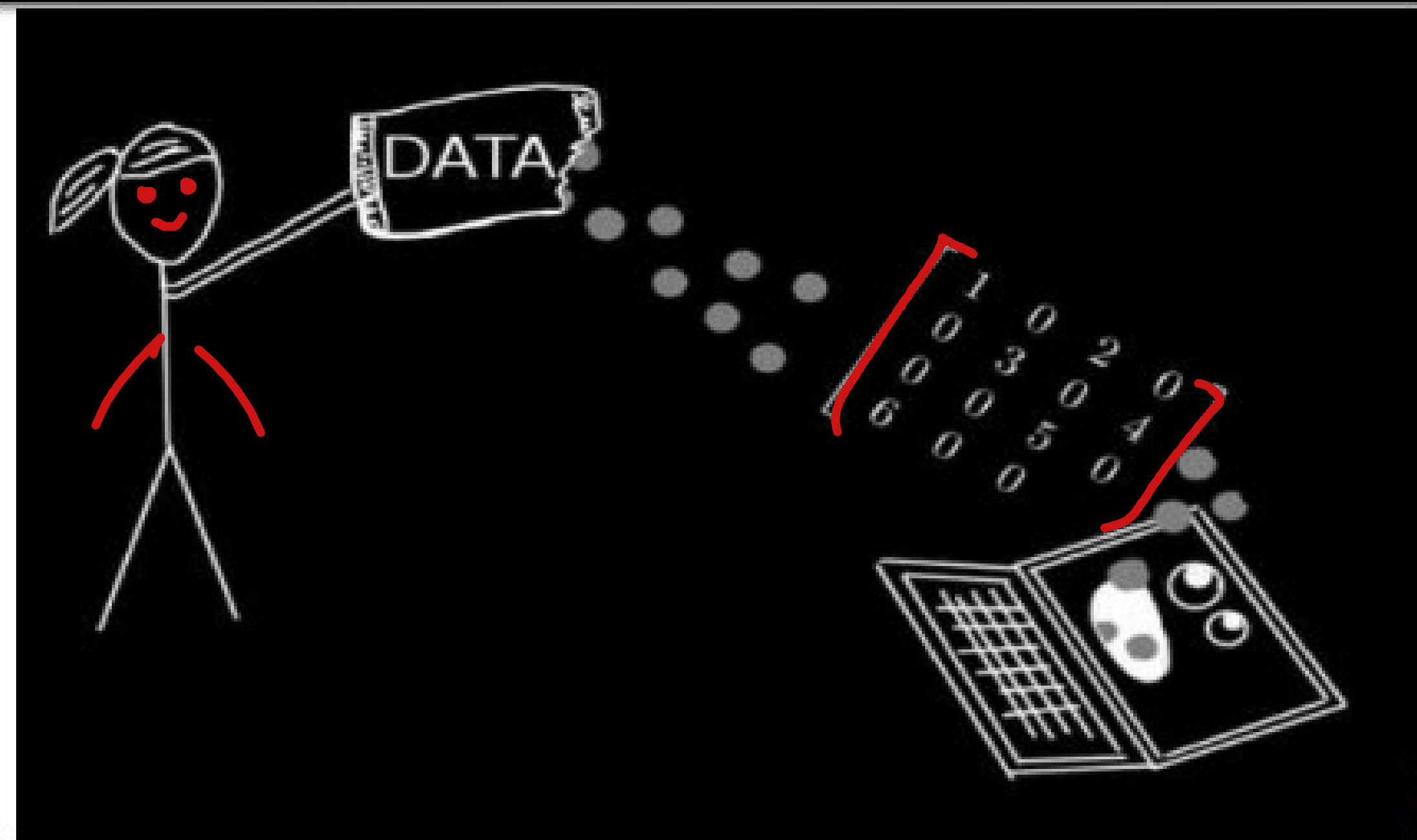
Why Import of Linear Algebra?

Linear Algebra is **fuel** of machine learning...

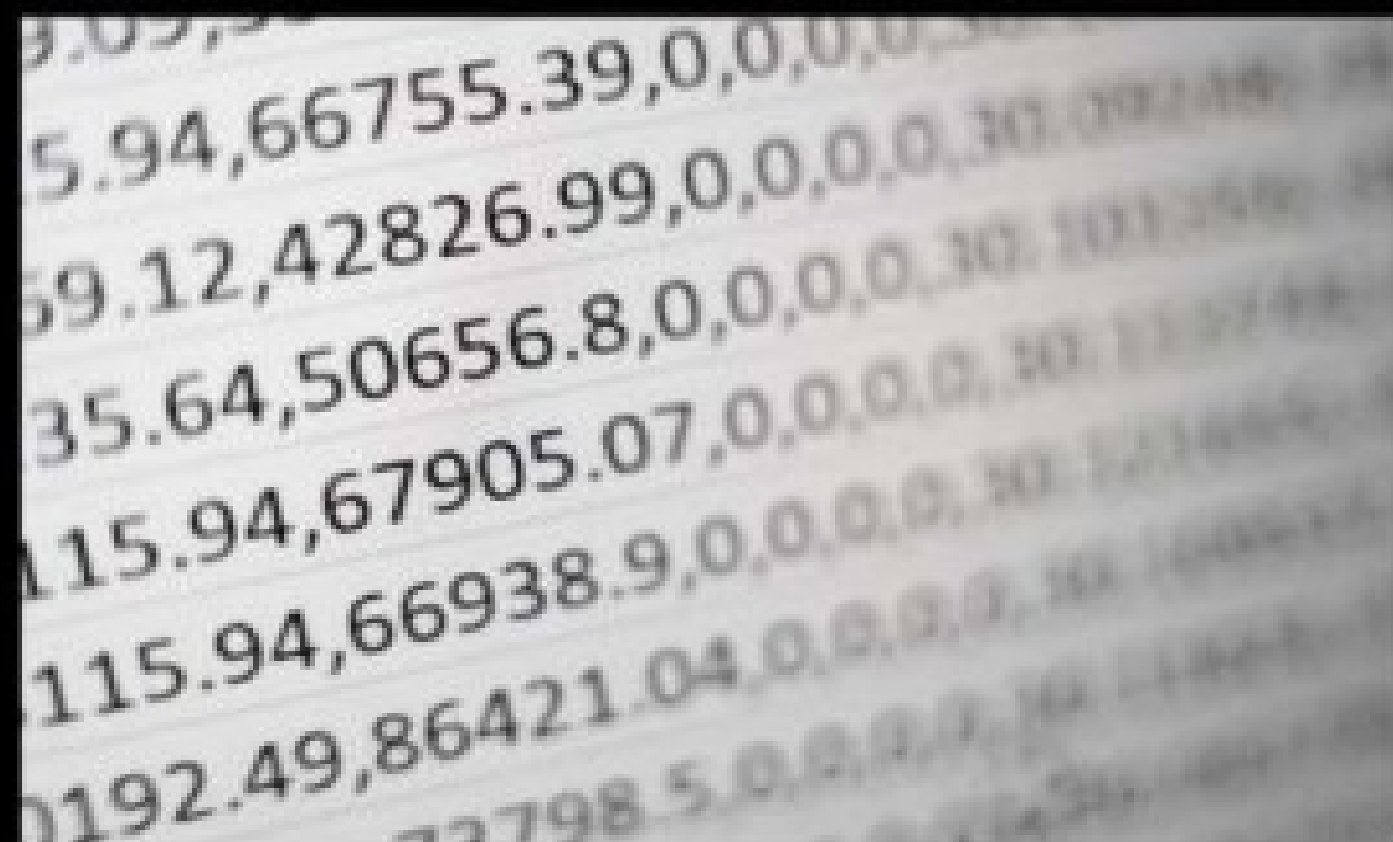
Every Type of data can be represented as matrices or as vectors.



$M - c. of R/c$



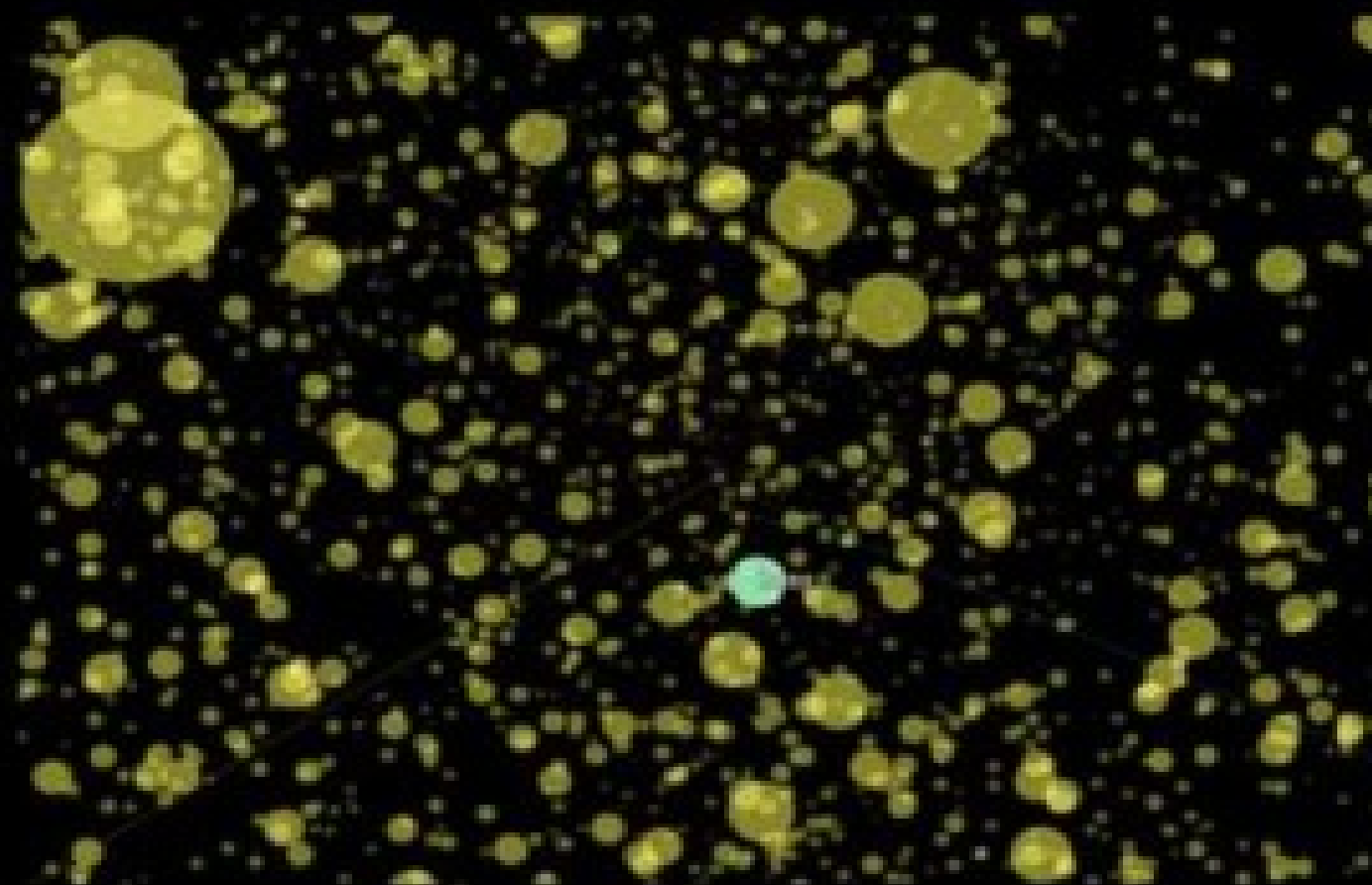
Common Areas of Application—Linear Algebra in Action



Data Representation

Using vectors, matrices and tensors to represent data.

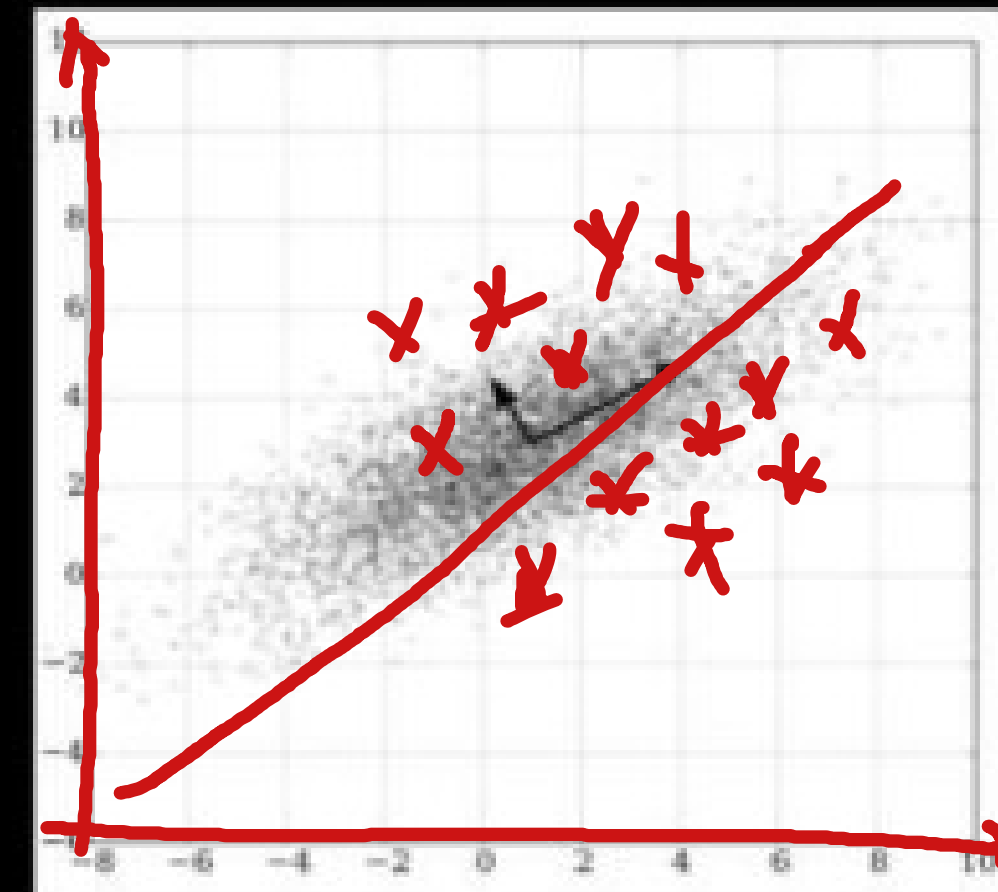
$\text{Data} \rightarrow M/V/T$



Word Embeddings

Embedding vectors to efficiently represent words for natural language problems

(NLP)



Dimensionality Reduction

Using eigenvectors and eigenvalues to deal with large-dimensional data.

$|A - \lambda X| = 0$





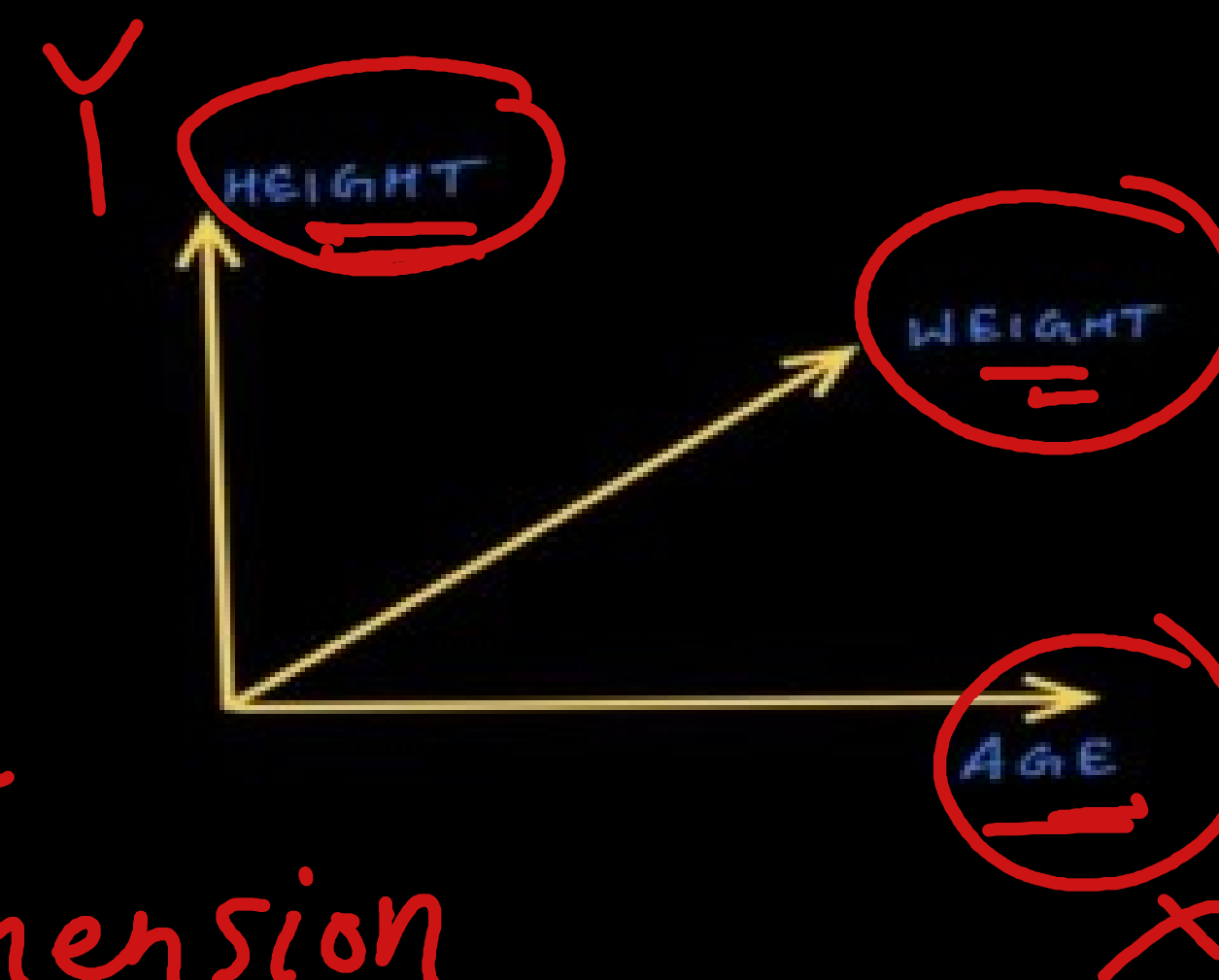
Data Representation

Height	Age	Weight
170	24	70
165	45	65
190	28	102
180	34	83
182	30	79
178	67	85

i/p

$\begin{bmatrix} 170 \\ 24 \\ 70 \end{bmatrix}$

Dimension

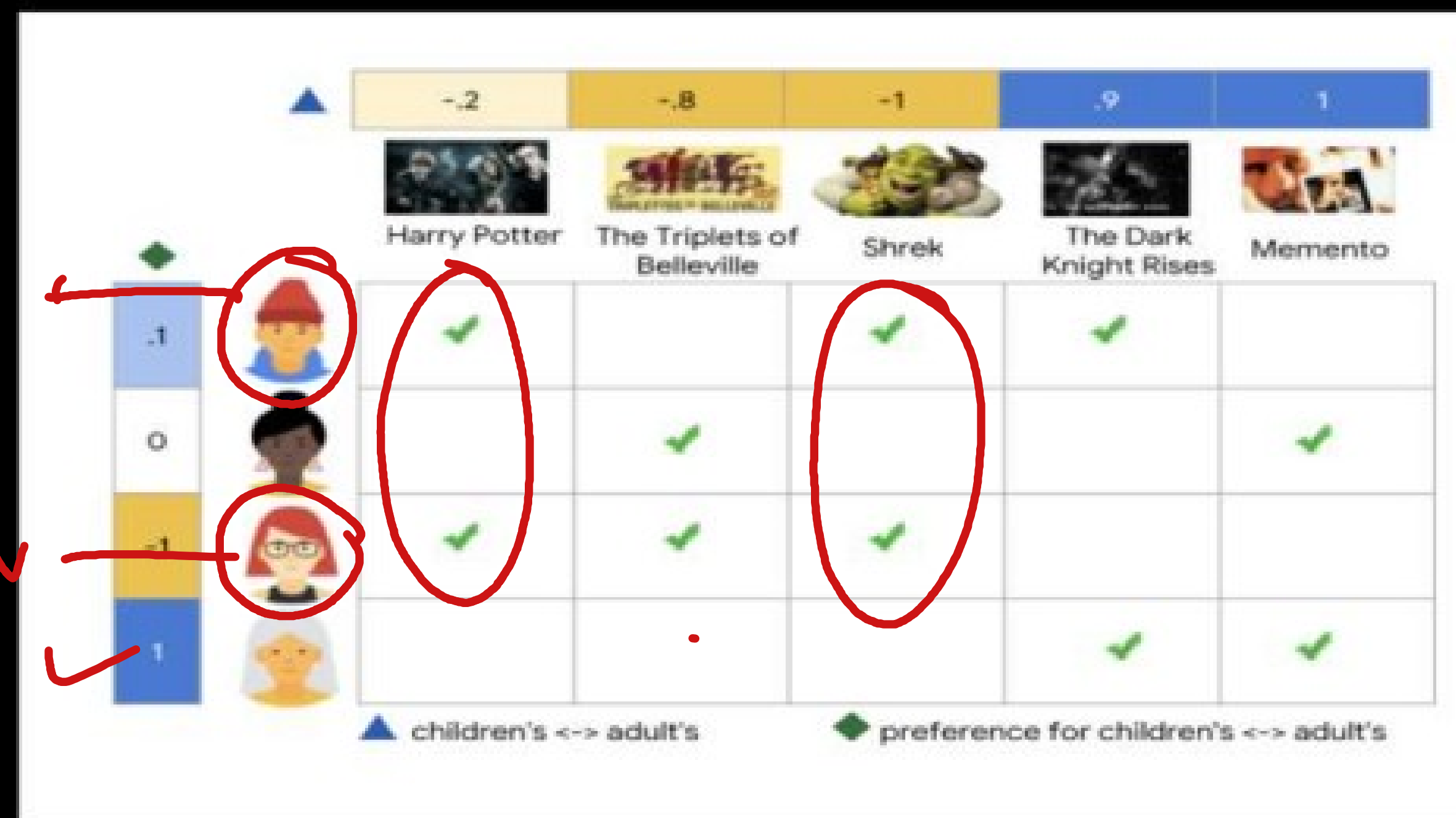


Recommendation Engines — Making use of embeddings

Movies recommendations

- Different types of users - kids, adults, teenagers, etc.
- Ratings of movies.
- Types of movies.
- Movies for kids and for adults.

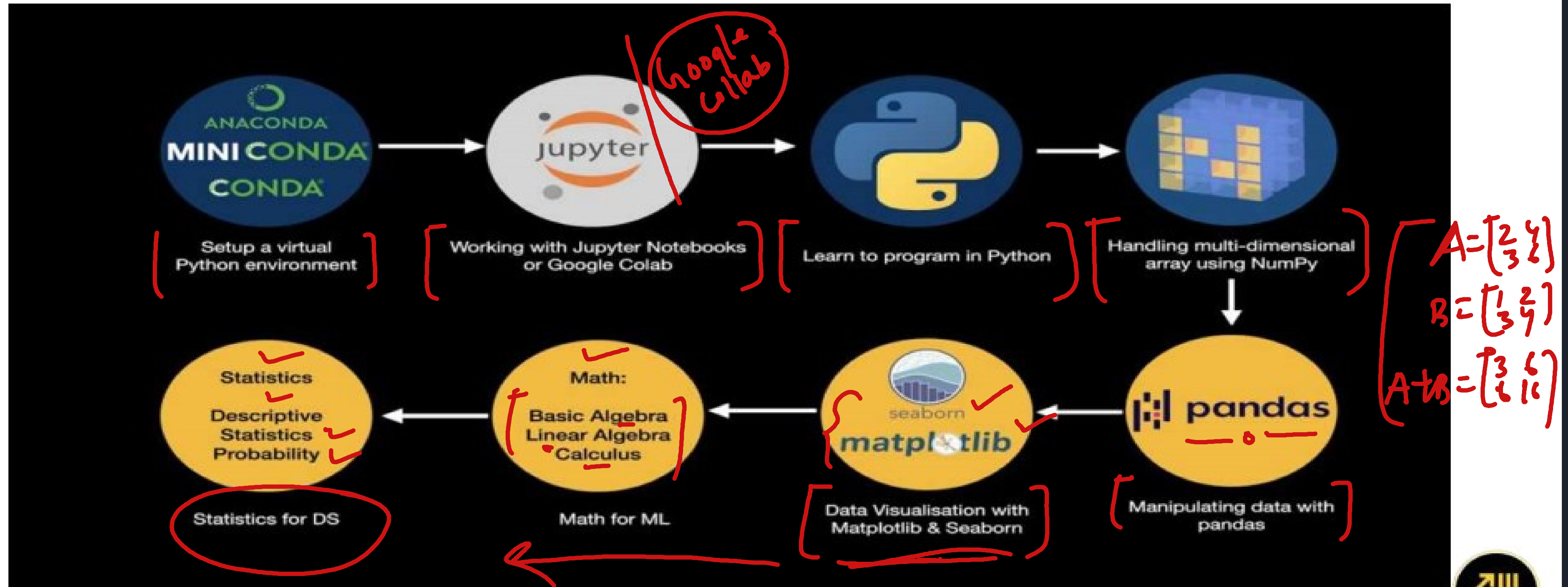
R-S — Algo



Source: Google's Machine Learning course on Recommendation Systems



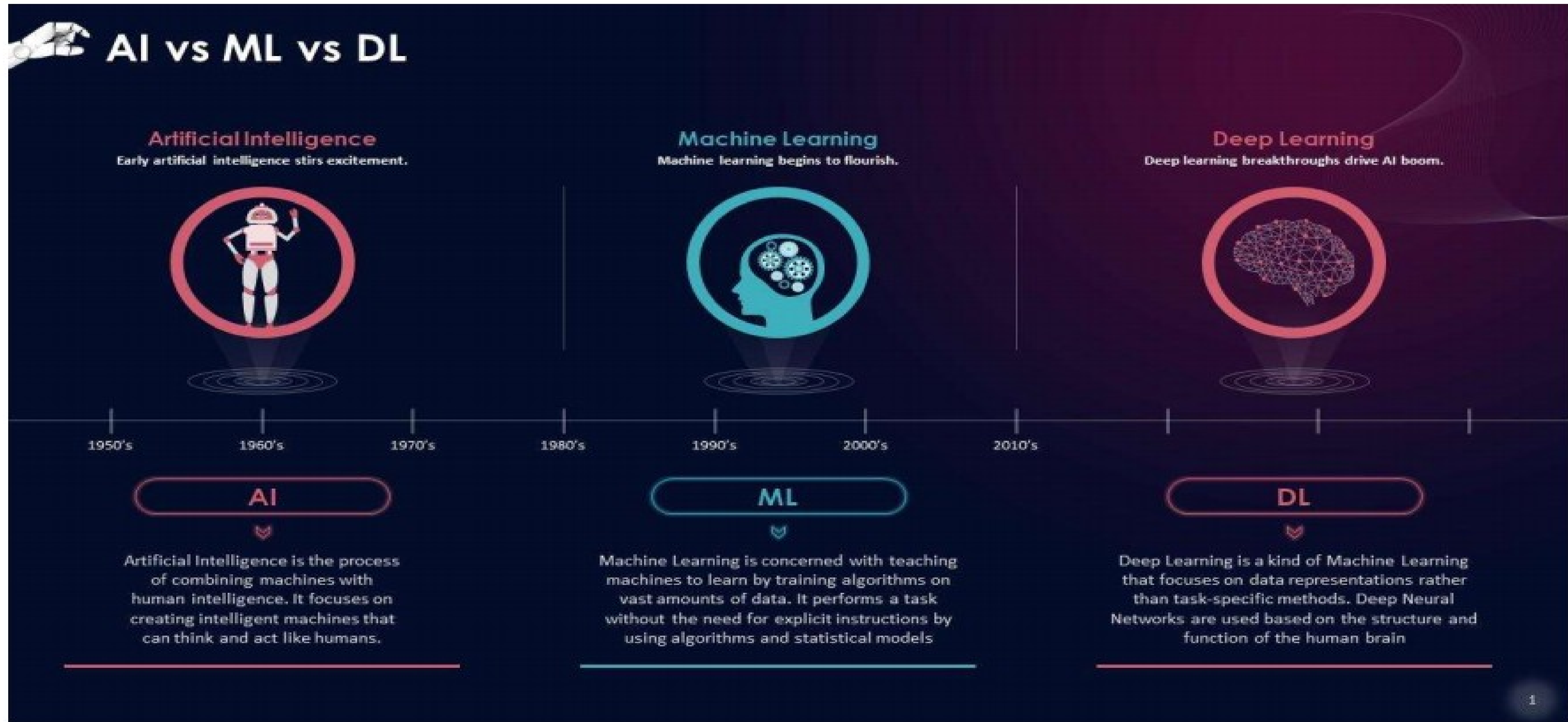
How Machine Learning Uses Linear Algebra to Solve Data Problems



Industries where Linear Algebra is used heavily

- ✓ Statistics
- ✓ Chemical Physics
- ✓ Genomics
- Word Embeddings – neural networks/deep learning
- ✓ Robotics (Arduinos, Math, Hardware (ME/EE))
- ✓ Image Processing
- ✓ Quantum Physics
- Cryptography





Recommended resources:-

Playlist on Linear Algebra by 3Blue1Brown **** — very engaging visualizations that explains the essence of linear algebra and its applications. Might be a little too hard for beginners. ✓✓

Khan Academy: Linear Algebra.

<https://www.khanacademy.org/math/linear-algebra> } ✓✓

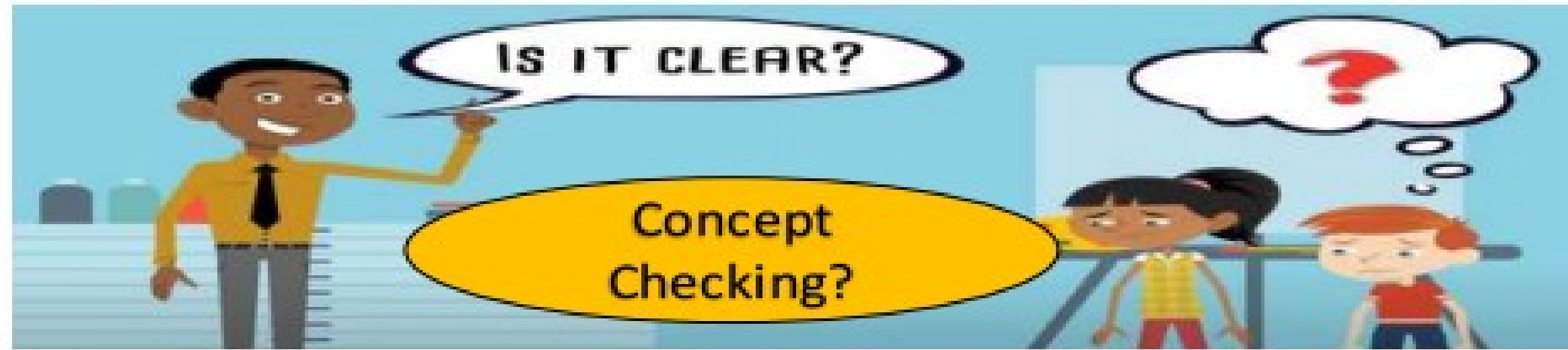
idea!



Foundations of AI & ML — You can follow and focused on my lectures that is enough to understanding of Programming, Math (Basic Algebra, Linear Algebra & Calculus) and Probability & Statistics. A complete Roadmap for first steps to learning DS/ML/AI.)

lecture + Problem solⁿ
+ Revision
+ Handwritten
note





Story of Linear Algebra :-

Story of determinant :

A. $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = \textcircled{ad - bc}$ ✓

B. $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = bc - ad$ ✗

C. $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ab - cd$ ✗

$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$ ✗

Linear Equation

✓ $2x + 3 = 4x$

✓ $x + 2y = 10$

✗ $\textcircled{x^2} + y = 100$

Not L.E

eq-6

$$x - y = 10 \quad (\text{L.E of two variable})$$

Solⁿ: A. Unique solⁿ

B. No solⁿ

C. ∞ solⁿ

$$x - y = 10$$

$$y = x - 10$$

$$y \mid x=0 = -10$$

$$y \mid x=10 = 0 \quad (0, 10)$$

$$(10, 0), (0, -10) \text{ --- } \infty$$

